

HHL quantum algorithm for solving a system of linear equations

Aram W. Harrow, Avinatan Hassidim, and Seth Lloyd, Phys. Rev. Lett. **103**, 150502 (2009)

Dr. Arun Sehrawat

video

Q_{AI} A system of linear equations

$N = \text{number of Equations} = \text{number of variables}$

Torale • We need to find x that satisfies

$$\underbrace{\begin{pmatrix} a_{11} & \cdots & a_{1N} \\ \vdots & \ddots & \vdots \\ a_{N1} & \cdots & a_{NN} \end{pmatrix}}_{\substack{A \\ \text{given}}} \underbrace{\begin{pmatrix} x_1 \\ \vdots \\ x_N \end{pmatrix}}_{\substack{x \\ \text{var}}} = \underbrace{\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_N \end{pmatrix}}_{\substack{b \\ \text{given}}}$$

• If A^{-1} exists, then $x = A^{-1}b$.

$\lambda_j \neq 0$
 $\det(A) \neq 0$

↑
solution

• If A is not a Hermitian matrix, $A^\dagger \neq A$, then we take

$$\underbrace{\begin{pmatrix} 0_{N \times N} & A \\ A^\dagger & 0_{N \times N} \end{pmatrix}}_{H_A} \underbrace{\begin{pmatrix} 0_{N \times 1} \\ x \end{pmatrix}}_{(2N) \times 1} = \underbrace{\begin{pmatrix} b \\ 0 \end{pmatrix}}_{(2N) \times 1}$$

Hamiltonian encoding

$$\underbrace{\begin{pmatrix} 0_{N \times N} & (A^\dagger)^{-1} \\ A^{-1} & 0_{N \times N} \end{pmatrix}}_{H_A^{-1}} \underbrace{\begin{pmatrix} b \\ 0 \end{pmatrix}}_{(2N) \times 1} = \underbrace{\begin{pmatrix} 0_{N \times 1} \\ x \end{pmatrix}}_{(2N) \times 1}$$

$$H_A H_A^{-1} = I_{(2N) \times (2N)} = H_A^{-1} H_A$$

Example: $N=2$

$$\underbrace{\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}}_A \underbrace{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}}_x = \underbrace{\begin{pmatrix} 4 \\ 5 \end{pmatrix}}_b$$

$$\det(A) = -2 \neq 0$$

$$A A^{-1} = I = A^{-1} A$$

$\text{poly}(N)$ steps

$$\underbrace{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}}_x = \underbrace{\frac{1}{-2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}}_{A^{-1}} \underbrace{\begin{pmatrix} 4 \\ 5 \end{pmatrix}}_b = \frac{1}{2} \begin{pmatrix} -6 \\ 7 \end{pmatrix}$$

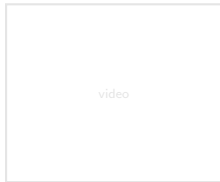
solution

$$N = 1 \text{ billion} = 10^9$$

$$\log_2(10^9) \approx \underline{\underline{30}} \text{ steps}$$

$$N = 2^n \leftarrow \text{number of qubits}$$

$$2N = 2^{n+1}$$



The HHL quantum algorithm

$$b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \quad \|b\| = \sqrt{|b_1|^2 + \dots + |b_n|^2}$$

norm

$$\lambda_j \neq 0$$

• We assume A is a Hermitian ($A^\dagger = A$) and non-singular (A^{-1} exists) matrix.

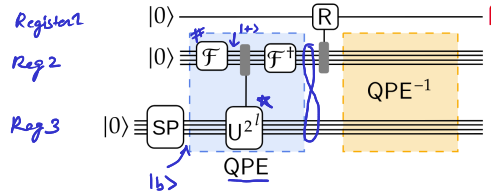
• Task: we want to realize $x = A^{-1}b$ with a quantum computer.

$$N < 2^n$$

Step 1: Prepare n qubits in the state $|b\rangle = \frac{1}{\|b\|} \sum_{i=1}^N b_i |i\rangle$ (amplitude encoding), where $N = 2^n$.
number of qubits

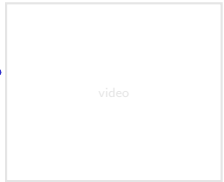
Change of basis $|i\rangle = \sum_{j=1}^N |u_j\rangle \langle u_j|i\rangle$ provides $|b\rangle = \sum_{j=1}^N \underbrace{\sum_{i=1}^N \frac{b_i}{\|b\|} \langle u_j|i\rangle}_{\beta_j} |u_j\rangle$, where $A|u_j\rangle = \lambda_j |u_j\rangle$.
cigenvalue
cigenkets
 $\langle u_j|u_{j'}\rangle = 0$ if $j \neq j'$
 $= 1$ otherwise

Step 2: Quantum Phase Estimation $|0\rangle|b\rangle \rightarrow \sum_{j=1}^N \beta_j |\lambda_j\rangle |u_j\rangle$, where $U = \exp(i2\pi A) = \sum_{j=1}^N e^{i2\pi \lambda_j} |u_j\rangle \langle u_j|$
eigen val
cigen vec of A
generated by A
Phase



$$\# \quad \sqrt{\frac{2}{T}} \sum_{t=0}^{T-1} \sin\left(\frac{\pi(t+\frac{1}{2})}{T}\right) |c\rangle$$

$$\star \quad \sum_{t=0}^{T-1} |TxT\rangle \otimes \cos\left(\frac{\pi t}{T}\right) |c\rangle$$



Q_{AI} The HHL quantum algorithm

$$\lambda_{\max} = \max_j \{ \lambda_j \}$$

$$\lambda_{\min} = \min_j \{ \lambda_j \}$$

Step 3: Conditional rotation $|0\rangle \sum_{j=1}^N \beta_j |\lambda_j\rangle |u_j\rangle \rightarrow \sum_{j=1}^N \left(\sqrt{1 - \left(\frac{c}{\lambda_j}\right)^2} |0\rangle + \frac{c}{\lambda_j} |1\rangle \right) |\lambda_j\rangle \beta_j |u_j\rangle$, $c \approx O\left(\frac{\lambda_{\min}}{\lambda_{\max}}\right)$

unentangle 2nd Reg.

Step 4: Inverse QPE

$$\rightarrow \sum_{j=1}^N \left(\sqrt{1 - \left(\frac{c}{\lambda_j}\right)^2} |0\rangle + \frac{c}{\lambda_j} |1\rangle \right) |0\rangle \beta_j |u_j\rangle$$

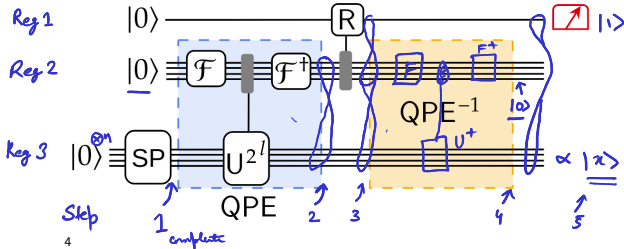
we want $\left(\frac{c}{\lambda_j}\right)^2 \leq 1$ for every j

Step 5: Measurement

$$\rightsquigarrow \sum_{j=1}^N \underbrace{\frac{c}{\lambda_j} \beta_j |u_j\rangle}_{\propto |b\rangle}$$

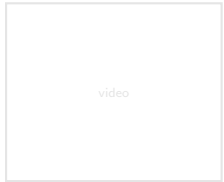
$c A^{-1} |b\rangle \propto |x\rangle$
not normalized ket

$$|b\rangle = \sum_{j=1}^N \beta_j |u_j\rangle$$



$$QPE = F U F^\dagger$$

$$(QPE)^\dagger = (F U F^\dagger)^\dagger = F U^\dagger F^\dagger$$





The HHL quantum algorithm (fine details)

$$\frac{1}{\kappa^2} \leq (\lambda_j)^2 \leq 1$$

Assumptions:

- 1 - The singular values of A lie between $\frac{1}{\kappa}$ and 1, that is, $\frac{1}{\kappa^2} I \leq A^\dagger A \leq I$, where $\kappa := \frac{\lambda_{\max}}{\lambda_{\min}}$ is **not too big**.
- 2 - A is s sparse (has at most s nonzero entries per row), then $U = \exp(i2\pi A)$ takes $O(\log(N) s^2)$ time. $s \ll N$

if A has no zero entry then $s = N$

- Under the assumptions, the HHL algorithm takes $O(\log(N) \kappa^2 \frac{1}{\epsilon})$ steps to provide a state in an ϵ -neighborhood of $|x\rangle$. If κ and $\frac{1}{\epsilon}$ are poly $\log(N)$, then the HHL algorithm provides the **exponential speed-up** in comparison to its classical counterpart. ⑥

$$x = A^{-1}b$$

length² of the vector $Ax - b$

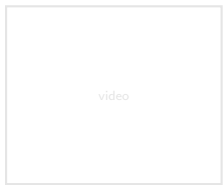
- One of the best classical matrix inversion algorithms, the conjugate gradient descent for $\min_x \|Ax - b\|^2$, take $O(N s \kappa \log(\frac{1}{\epsilon}))$ time for a general Hermitian matrix A .
2ⁿ at most total number of nonzero entries in A

number of comp in x

- Reading out all the components of $|x\rangle$ requires $N = 2^n$ steps. However, often one is interested in some expectation value $\langle x | M | x \rangle$. So, the exponential speed-up is about $\langle x | M | x \rangle$.

- Step 1, State Preparation (SP), $|0\rangle \rightarrow \frac{1}{\|b\|} \sum_{i=1}^N b_i |i\rangle$ may take $O(N)$ steps. Q-RAM

A. W. Harrow, A. Hassidim, and S. Lloyd, Phys. Rev. Lett. **103**, 150502 (2009).
 J. R. Shewchuk, Technical Report No. CMU-CS-94-125, Carnegie Mellon University, March 1994.
 M. Schuld and F. Petruccione, Supervised Learning with Quantum Computers (Springer, 2018), Chapter 3.



Data Loading : State Preparation

Step 1: $|b\rangle \rightarrow |b\rangle = \frac{1}{\|b\|} \sum_{i=0}^{N-1} b_i |i\rangle$

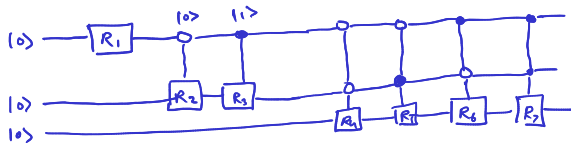
$$|b\rangle = \frac{1}{\sqrt{b_0^2 + b_1^2 + b_2^2 + b_3^2}} [b_0 |00\rangle + b_1 |01\rangle + b_2 |10\rangle + b_3 |11\rangle]$$

$n=2, N=2^n=4$

$$= \frac{1}{\|b\|} \left[\underbrace{\sqrt{b_2^2 + b_3^2}}_{\text{---}} |0\rangle \left[\frac{b_0 |0\rangle + b_1 |1\rangle}{\sqrt{b_0^2 + b_1^2}} \right] + \underbrace{\sqrt{b_2^2 + b_3^2}}_{\text{---}} |1\rangle \left[\frac{b_2 |0\rangle + b_3 |1\rangle}{\sqrt{b_2^2 + b_3^2}} \right] \right]$$

$$b = \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \end{bmatrix} \in \mathbb{R}^4$$

$$\frac{\sqrt{b_0^2 + b_1^2} |0\rangle + \sqrt{b_2^2 + b_3^2} |1\rangle}{\|b\|}$$



number of gates = $N-1 = O(N)$

" " qubits = $\log(N) = n$

$$R_1 = \frac{1}{\|b\|} \begin{bmatrix} \sqrt{b_0^2 + b_1^2} & -\sqrt{b_2^2 + b_3^2} \\ \sqrt{b_2^2 + b_3^2} & \sqrt{b_0^2 + b_1^2} \end{bmatrix}$$

$$R_2 = \begin{bmatrix} b_0 & -b_1 \\ b_1 & b_0 \end{bmatrix} \frac{1}{\sqrt{b_0^2 + b_1^2}}$$

$$R_3 = \begin{bmatrix} b_2 & -b_3 \\ b_3 & b_2 \end{bmatrix} \frac{1}{\sqrt{b_2^2 + b_3^2}}$$

Rotation around y axis

$$\sum_{i=0}^{N-1} [\cos(b_i) |0\rangle + \sin(b_i) |1\rangle]$$

number of qubits = N
" " gates = N



video

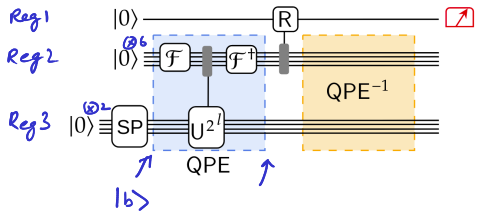
The HHL quantum algorithm: an example

$$A = \frac{1}{6} \left(\begin{array}{ccc|c} 2 & 2 & -1 & 0 \\ 2 & 3 & -2 & 0 \\ -1 & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)_{4 \times 4}, \quad \{\lambda_i\} = \frac{1}{6} \{3 \pm 2\sqrt{2}, 1, 0\} \approx \{0.111110, 0.000001, 0.001010, 0.000000\}_{\text{Binary}}$$

$N = 4 = 2^n, n = 2$
 $0 \leq \lambda_j < 1$
 $A|u_j\rangle = \lambda_j|u_j\rangle$
 $|u_4\rangle = |11\rangle$
 $\lambda_4 = 0$

Step 1: State Preparation $|b\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |10\rangle) = \frac{1}{\sqrt{6}}(-|u_1\rangle + |u_2\rangle + 2|u_3\rangle)$

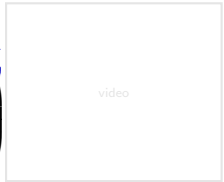
Step 2: Quantum Phase Estimation $|0\rangle^{\otimes 6} |b\rangle \rightarrow \frac{1}{\sqrt{6}}(-|111110\rangle|u_1\rangle + |000001\rangle|u_2\rangle + 2|001010\rangle|u_3\rangle)$



Classical comp

$$\begin{pmatrix} 12 & -12 & -6 & 0 \\ -12 & 18 & 12 & 0 \\ -6 & 12 & 12 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -6 \\ 18 \\ 18 \\ 0 \end{pmatrix}$$

$x_{\text{act}} = \begin{pmatrix} -6 \\ 18 \\ 18 \\ 0 \end{pmatrix}$





The HHL quantum algorithm: an example

$$\frac{c}{\lambda_1} = \frac{6c}{3+2\sqrt{2}}$$

$$\frac{1}{\lambda_2} = \frac{6}{3-2\sqrt{2}}$$

Step 3: Conditional rotation

$$|0\rangle \frac{1}{\sqrt{6}} \left(-|111110\rangle|u_1\rangle + |000001\rangle|u_2\rangle + 2|001010\rangle|u_3\rangle \right) \rightarrow$$

$$-\frac{1}{\sqrt{6}} \left(\sqrt{1 - \left(\frac{6c}{3+2\sqrt{2}} \right)^2} |0\rangle + \frac{6c}{3+2\sqrt{2}} |1\rangle \right) |111110\rangle|u_1\rangle + \frac{1}{\sqrt{6}} \left(\sqrt{1 - \left(\frac{6c}{3-2\sqrt{2}} \right)^2} |0\rangle + \frac{6c}{3-2\sqrt{2}} |1\rangle \right) |000001\rangle|u_2\rangle$$

$$+ \frac{2}{\sqrt{6}} \left(\sqrt{1 - (6c)^2} |0\rangle + 6c |1\rangle \right) |001010\rangle|u_3\rangle$$

$$c = \frac{1}{2} \left(\frac{3-2\sqrt{2}}{3+2\sqrt{2}} \right) = 0 \left(\frac{\lambda_{min}}{\lambda_{max}} \right)$$

Step 4: Inverse QPE $\rightarrow$

$$-\frac{1}{\sqrt{6}} \left(\sqrt{1 - \left(\frac{6c}{3+2\sqrt{2}} \right)^2} |0\rangle + \frac{6c}{3+2\sqrt{2}} |1\rangle \right) |0\rangle|u_1\rangle + \frac{1}{\sqrt{6}} \left(\sqrt{1 - \left(\frac{6c}{3-2\sqrt{2}} \right)^2} |0\rangle + \frac{6c}{3-2\sqrt{2}} |1\rangle \right) |0\rangle|u_2\rangle$$

$$+ \frac{2}{\sqrt{6}} \left(\sqrt{1 - (6c)^2} |0\rangle + 6c |1\rangle \right) |0\rangle|u_3\rangle$$

Step 5: Measurement $\rightsquigarrow |1\rangle \left(\underbrace{\frac{6c}{3+2\sqrt{2}}}_{\frac{c}{\lambda_1}} \underbrace{\frac{(-1)}{\sqrt{6}}}_{\beta_1} |u_1\rangle + \underbrace{\frac{6c}{3-2\sqrt{2}} \frac{1}{\sqrt{6}}}_{c A^{-1}|\mathbf{b}\rangle} |u_2\rangle + \underbrace{6c \frac{2}{\sqrt{6}}}_{\frac{c}{\lambda_3} \beta_3} |u_3\rangle \right)$

$$|\mathbf{b}\rangle = \sum_{j=1}^n \beta_j |u_j\rangle$$

$$\beta_4 = 0$$

